

Understanding how PlatformIO manages libraries and toolchains for development

PlatformIO

- It is a open source development environment.
- It is cross-platform,cross-architecture for writing code to microcontrollers like STM32, Arduino.
- It automatically integrates compilers, toolchains, tags necessary dependencies and libraries so we don't have to do it manually.

Problem: complexity of setting up and managing. Different toolchains and IDEs required for different microcontrollers and boards.

PlatformIO Core: command line interface of platformIO . The other PlatformIO-based software and IDEs are based on **PlatformIO Core CLI**, such as PlatformIO IDE. Works on any system with python. Used for more advanced users and automation while IDE is more beginner-friendly .

Managing libraries

PlatformIO manages libraries through its PlatformIO library manager. It is a tool which manages libraries of PlatformIO Registry and searches for them.

It searches for libraries in the registry.

It installs libraries and their dependencies.

It manages versions and updates the library when a new version is available.

Two ways to store and manage libraries:

- **Per-Project Libraries** (specific to a single project): Library are present inside the project folder (.pio/libdeps/) – enables portability of project as all libraries remains intact along with the code. More space usage.
- **Global Libraries** (shared across multiple projects): Installs globally. Saves space by allowing multiple projects to use the same library. Beneficial when many projects are using the same library. Global libraries are present in the PlatformIO's global directory. Saves space.

library.json is a manifest file of a library package. It allows developers to keep a project in its own structure.

Fields in the library.json manifest file used in PlatformIO to describe and configure a library:

1 **\$schema** → Defines the schema (format rules) for validation.

2 **name** → The name of the library.

3 **version** → The version number (e.g., "1.0.0").

4 **description** → A short description of what the library does.

5 **keywords** → Tags to help find the library (e.g., ["sensor", "motor"]).

6 **homepage** → The official website or documentation link.

7 **repository** → A link to the library's **GitHub/GitLab** repo.

8 **authors** → The creator(s) of the library.

9 **license** → Defines the usage rights (e.g., "MIT" or "GPL-3.0").

Managing toolchain

A toolchain is a set of tools (compiler, linker, debugger, etc.) needed to build and upload code for a microcontroller (MCU).

When a project is created, platformio.ini file is read and the necessary toolchains are installed. By default, latest version is downloaded but version can be specified in the configuration file. Working with multiple MCU's, PlatformIO will **automatically switch toolchains** based on your project.

Platformio.ini is the configuration file for platformio projects. It tells PlatformIO which board, framework, libraries, and toolchains to use.